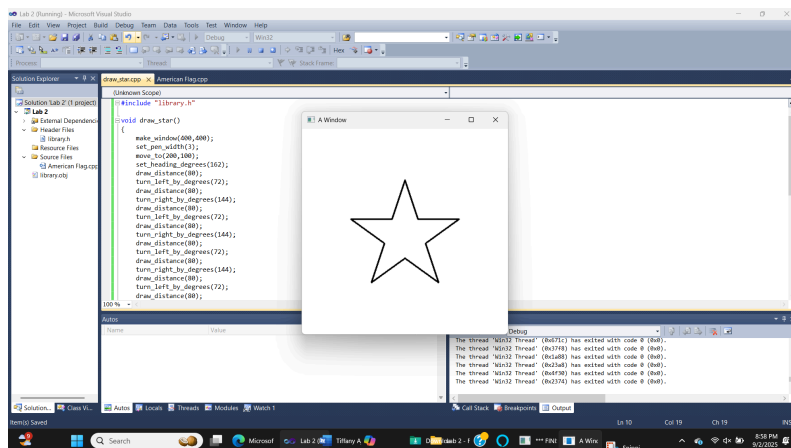


1. A five pointed star

```
#include "library.h"
```

```
void main()
{
    make_window(400,400);
    set_pen_width(3);
    move_to(200,100);
    set_heading_degrees(162);
    draw_distance(80);
    turn_left_by_degrees(72);
    draw_distance(80);
    turn_right_by_degrees(144);
    draw_distance(80);
    turn_left_by_degrees(72);
    draw_distance(80);
    turn_right_by_degrees(144);
    draw_distance(80);
    turn_left_by_degrees(72);
    draw_distance(80);
    turn_right_by_degrees(144);
    draw_distance(80);
    turn_left_by_degrees(72);
    draw_distance(80);
    turn_right_by_degrees(144);
    draw_distance(80);
    turn_left_by_degrees(18);
}
```



2. A function

```
#include "library.h"
```

```
void draw_star(const int x, const int y, const int length)
{
```

```
    start_shape();
    set_heading_degrees(162);
    draw_distance(length);
    turn_left_by_degrees(72);
    draw_distance(length);
    turn_right_by_degrees(144);
    draw_distance(length);
    turn_left_by_degrees(72);
    draw_distance(length);
    turn_right_by_degrees(144);
    draw_distance(length);
    turn_left_by_degrees(72);
    draw_distance(length);
    turn_right_by_degrees(144);
    draw_distance(length);
    turn_left_by_degrees(72);
    draw_distance(length);
    turn_right_by_degrees(144);
    draw_distance(length);
    turn_left_by_degrees(72);
    draw_distance(length);
    turn_left_by_degrees(18);
```

```
}
```

```
void main()
```

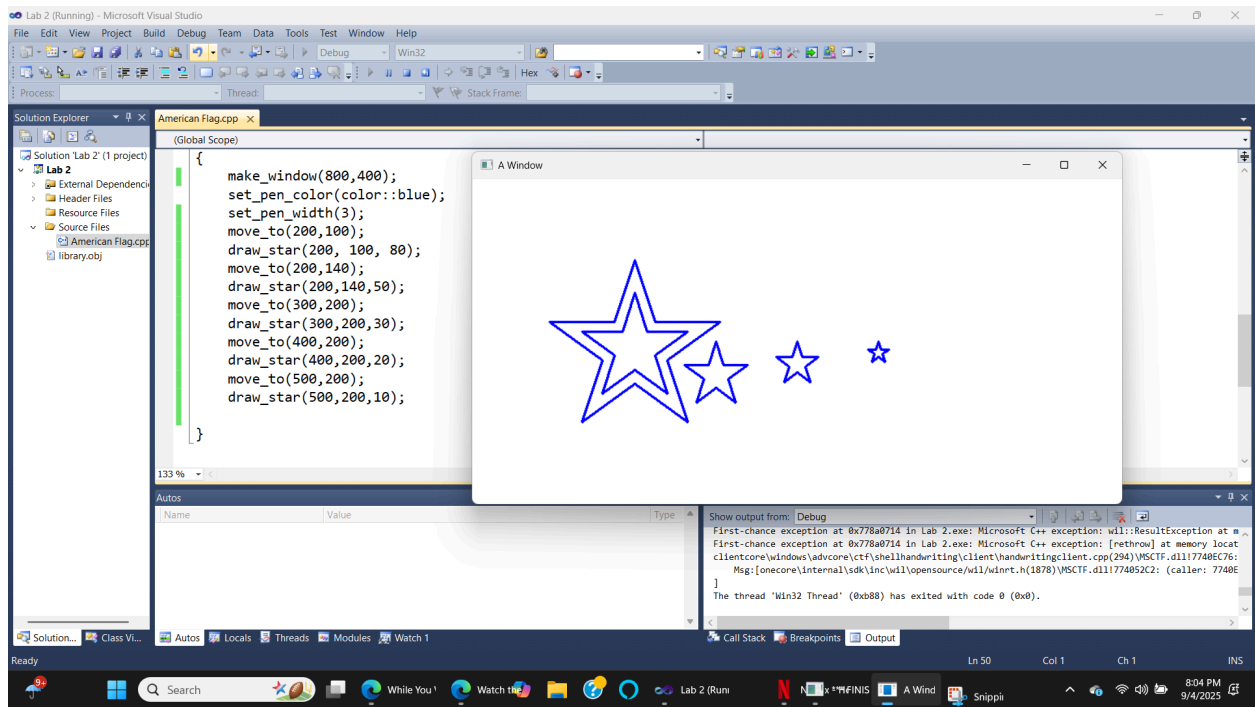
```
{
```

```
    make_window(800,400);
    set_pen_color(color::blue);
    set_pen_width(3);
    move_to(200,100);
    draw_star(200, 100, 80);
    move_to(200,140);
    draw_star(200,140,50);
    move_to(300,200);
```

```

draw_star(300,200,30);
move_to(400,200);
draw_star(400,200,20);
move_to(500,200);
draw_star(500,200,10);
}

```



3. Coloring it in

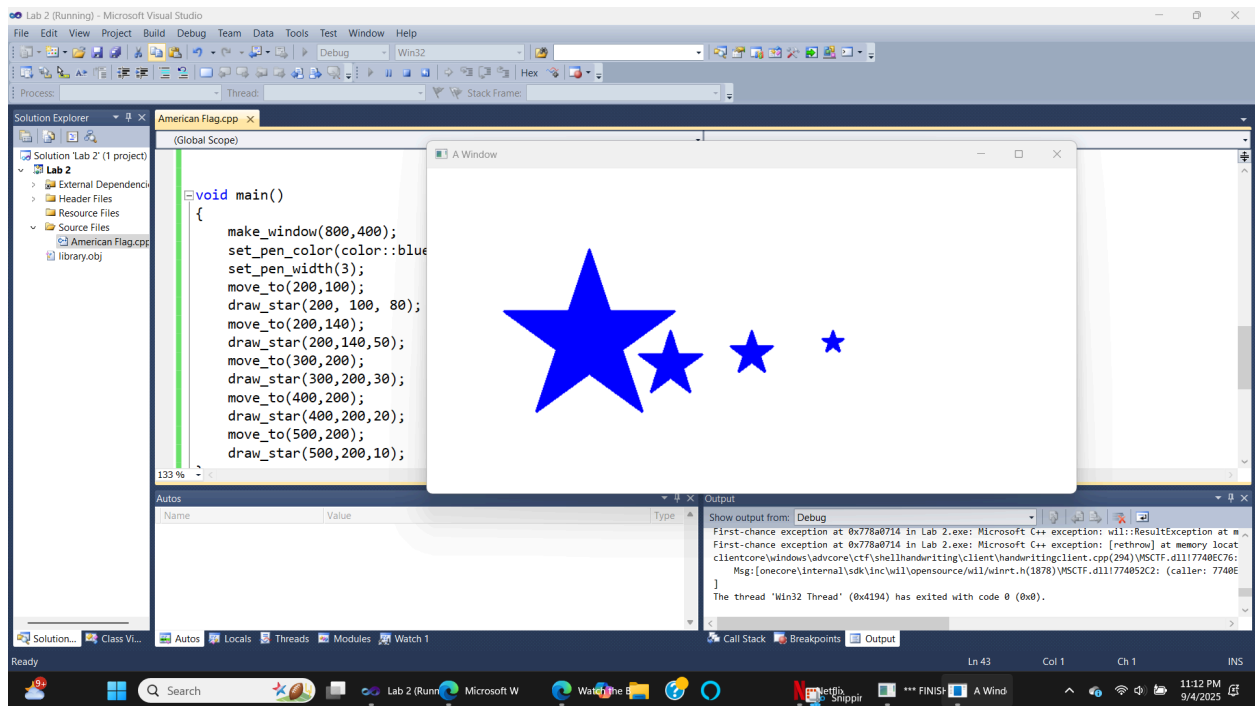
```
#include "library.h"
```

```
void draw_star(const int x, const int y, const int length)
{
    start_shape();
    set_heading_degrees(162);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_left_by_degrees(18);
    set_pen_color(color::blue);
    fill_shape();
}
```

```

void main()
{
    make_window(800,400);
    set_pen_color(color::blue);
    set_pen_width(3);
    move_to(200,100);
    draw_star(200, 100, 80);
    move_to(200,140);
    draw_star(200,140,50);
    move_to(300,200);
    draw_star(300,200,30);
    move_to(400,200);
    draw_star(400,200,20);
    move_to(500,200);
    draw_star(500,200,10);
}

```



4. The Lone-Star Program

```
#include "library.h"

void draw_star(const int x, const int y, const int length)
{
    move_to(x,y);
    start_shape();
    set_heading_degrees(162);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_left_by_degrees(18);
    fill_shape();
}
```

```

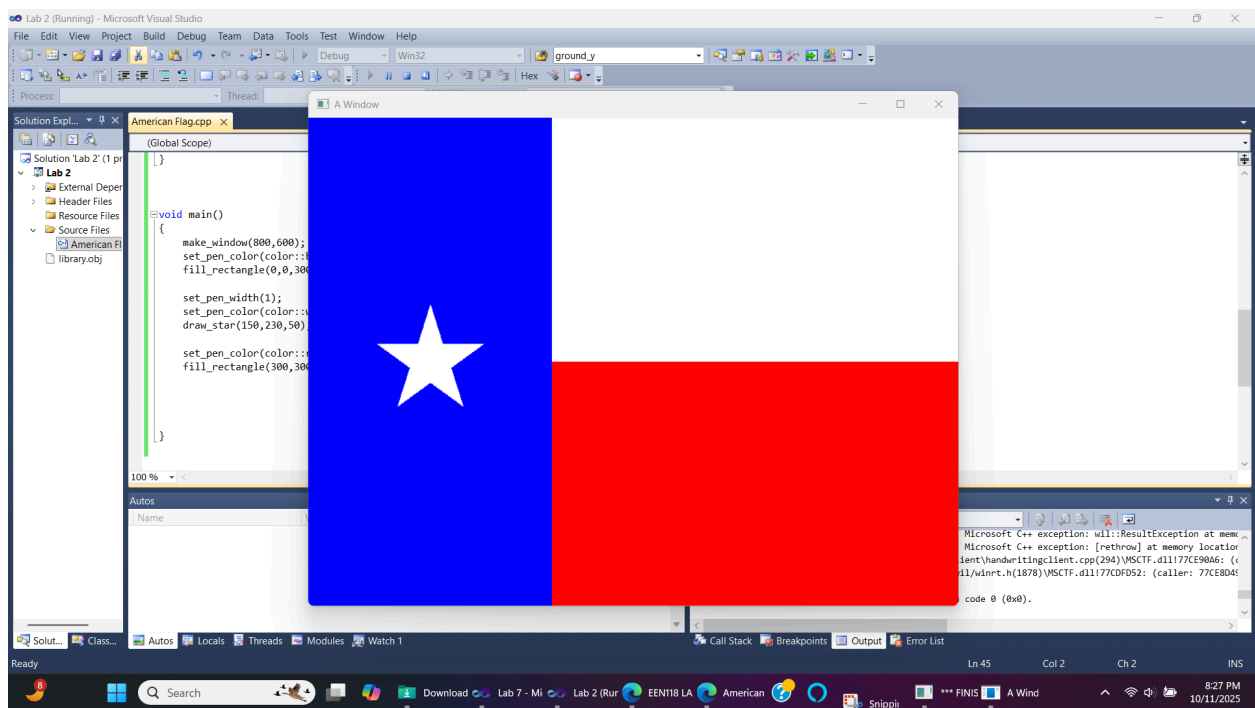
void main()
{
    make_window(800,600);
    set_pen_color(color::blue);
    fill_rectangle(0,0,300,600);

    set_pen_width(1);
    set_pen_color(color::white);
    draw_star(150,230,50);

    set_pen_color(color::red);
    fill_rectangle(300,300,800,300);

}

```



5. Row of stars

```
#include "library.h"

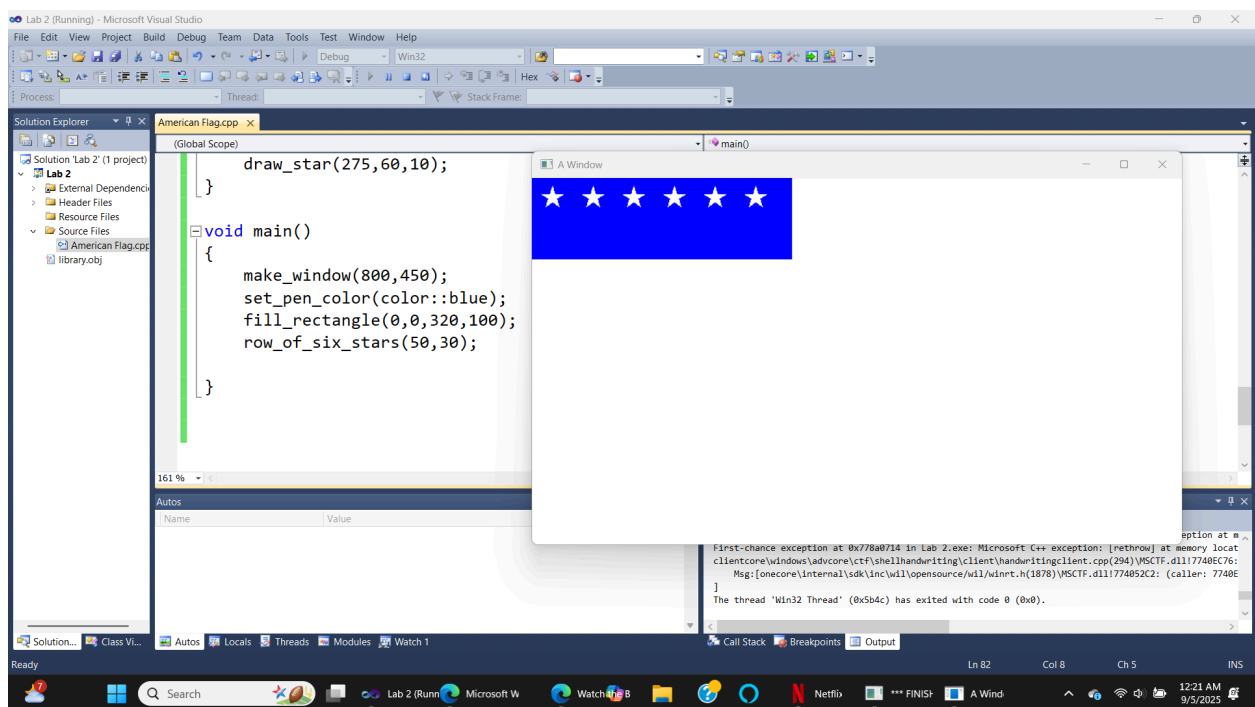
void draw_star(const int x, const int y, const int length)
{
    start_shape();
    set_heading_degrees(162);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_left_by_degrees(18);
    set_pen_color(color::white);
    fill_shape();
}
```

```

void row_of_six_stars(const int x, const int y, int scale)
{
    set_pen_color(color::white);
    set_pen_width(1);
    move_to(25,10);
    draw_star(25 * scale, 10 * scale, 10);
    move_to(75,10);
    draw_star(75 * scale, 10 * scale, 10);
    move_to(125,10);
    draw_star(125 * scale, 10 * scale, 10);
    move_to(175,10);
    draw_star(175 * scale, 10 * scale, 10);
    move_to(225,10);
    draw_star(225 * scale, 10 * scale, 10);
    move_to(275,10);
    draw_star(275 * scale, 10 * scale, 10);
}

void main()
{
    make_window(800,450);
    set_pen_color(color::blue);
    fill_rectangle(0,0,320,100);
    row_of_six_stars(50,30,1);
}

```



6. Blocks of stars

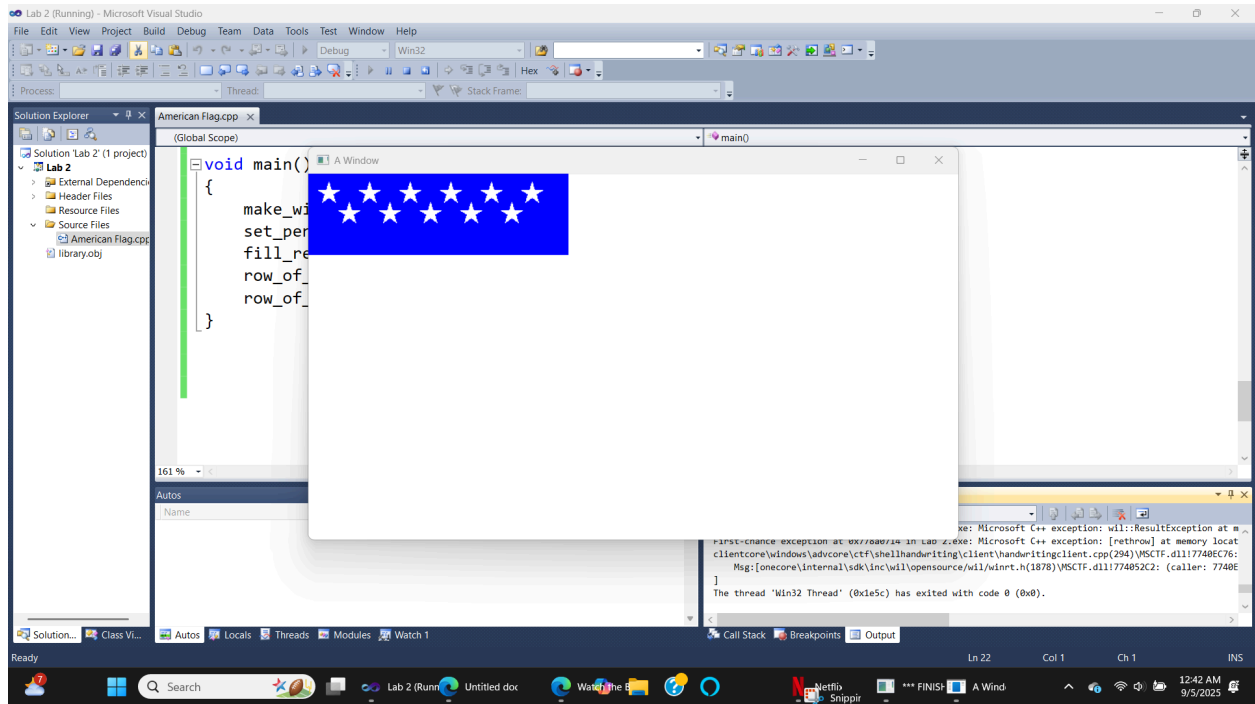
```
#include "library.h"
```

```
void draw_star(const int x, const int y, const int length)
{
    start_shape();
    set_heading_degrees(162);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_left_by_degrees(18);
    set_pen_color(color::white);
    fill_shape();
}
```

```
void row_of_six_stars(const int x, const int y, int scale)
{
    move_to(25,10);
    draw_star(25 * scale, 10 * scale, 10);
    move_to(75,10);
    draw_star(75 * scale, 10 * scale, 10);
    move_to(125,10);
    draw_star(125 * scale, 10 * scale, 10);
    move_to(175,10);
    draw_star(175 * scale, 10 * scale, 10);
    move_to(225,10);
    draw_star(225 * scale, 10 * scale, 10);
    move_to(275,10);
    draw_star(275 * scale, 10 * scale, 10);
}
```

```
void row_of_five_stars(const int x, const int y, int scale)
{
    move_to(50,35);
    draw_star(50 * scale, 35 * scale, 10);
    move_to(100,35);
    draw_star(100 * scale, 35 * scale, 10);
    move_to(150,35);
    draw_star(150 * scale, 35 * scale, 10);
    move_to(200,35);
    draw_star(200 * scale, 35 * scale, 10);
    move_to(250,35);
    draw_star(250 * scale, 35 * scale, 10);
}
```

```
void main()
{
    make_window(800,450);
    set_pen_color(color::blue);
    fill_rectangle(0,0,320,100);
    row_of_six_stars(50,30,1);
    row_of_five_stars(60,35,1);
}
```



7. The American Flag

```
#include "library.h"

void draw_star(const int x, const int y, const int length)
{
    start_shape();
    set_heading_degrees(162);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_right_by_degrees(144);
    move_distance(length);
    note_position();
    turn_left_by_degrees(72);
    move_distance(length);
    note_position();
    turn_left_by_degrees(18);
    set_pen_color(color::white);
    fill_shape();
}
```

```

void row_of_six_stars(const int y, int scale)
{
    move_to(25 * scale, y * scale);
    draw_star(25 * scale, y * scale, 10 * scale);
    move_to(75 * scale, y * scale);
    draw_star(75 * scale, y * scale, 10 * scale);
    move_to(125 * scale, y * scale);
    draw_star(125 * scale, y * scale, 10 * scale);
    move_to(175 * scale, y * scale);
    draw_star(175 * scale, y * scale, 10 * scale);
    move_to(225 * scale, y * scale);
    draw_star(225 * scale, y * scale, 10 * scale);
    move_to(275 * scale, y * scale);
    draw_star(275 * scale, y * scale, 10 * scale);
}

```

```

void row_of_five_stars(const int y, int scale)
{
    move_to(50 * scale, y * scale);
    draw_star(50 * scale, y * scale, 10 * scale);
    move_to(100 * scale, y * scale);
    draw_star(100 * scale, y * scale, 10 * scale);
    move_to(150 * scale, y * scale);
    draw_star(150 * scale, y * scale, 10 * scale);
    move_to(200 * scale, y * scale);
    draw_star(200 * scale, y * scale, 10 * scale);
    move_to(250 * scale, y * scale);
    draw_star(250 * scale, y * scale, 10 * scale);
}

```

```

void American_flag(int scale)
{
    make_window(800 * scale, 450 * scale);
    set_pen_color(color::blue);
    fill_rectangle(0, 0, 300 * scale, 220 * scale);
    row_of_six_stars(20, scale);
    row_of_five_stars(40, scale);
    row_of_six_stars(60, scale);
    row_of_five_stars(80, scale);
    row_of_six_stars(100, scale);
    row_of_five_stars(120, scale);
    row_of_six_stars(140, scale);
}

```

```

row_of_five_stars(160, scale);
row_of_six_stars(180, scale);

set_pen_color(color::red);
fill_rectangle(300 * scale, 0, 800 * scale, 30 * scale);
fill_rectangle(300 * scale, 65 * scale, 800 * scale, 30 * scale);
fill_rectangle(300 * scale, 125 * scale, 800 * scale, 30 * scale);
fill_rectangle(300 * scale, 190 * scale, 800 * scale, 30 * scale);
fill_rectangle(0, 260 * scale, 800 * scale, 30 * scale);
fill_rectangle(0, 340 * scale, 800 * scale, 30 * scale);
fill_rectangle(0, 420 * scale, 800 * scale, 30 * scale);
}

void main()
{
    American_flag(1);
}

```

